

Introduction to Machine Learning

Course code-CSE475

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DBSCAN Clustering Algorithm

Question: If Epsilon is .6 and minimum point is 4, what are the clusters that DBScan would discover with the following dataset.

Dataset

Item	Feature-1	Feature-2
A	1	2
B	3	4
C	2.5	4
D	1.5	2.5
E	3	5
F	2.8	4.5
G	2.5	4.5
H	1.2	2.5
I	1	3
J	1	5
K	1	2.5
L	5	6
M	4	3

Epsilon is .6

	A	B	C	D	E	F	G	H	I	J	K	L	M
A	0	8	6.25	0.5	13	9.49	8.5	0.29	1	9	0.25	32	10
B	8	0	0.25	4.5	1	0.29	0.5	5.49	5	5	6.25	8	2
C	6.25	0.25	0	3.25	1.25	0.34	0.25	3.94	3.25	3.25	4.5	10.25	3.25
D	0.5	4.5	3.25	0	8.5	5.69	5	0.09	0.5	6.5	0.25	24.5	6.5
E	13	1	1.25	8.5	0	0.29	0.5	9.49	8	4	10.25	5	5
F	9.49	0.29	0.34	5.69	0.29	0	0.09	6.56	5.49	3.49	7.24	7.09	3.69
G	8.5	0.5	0.25	5	0.5	0.09	0	5.69	4.5	2.5	6.25	8.5	4.5
H	0.29	5.49	3.94	0.09	9.49	6.56	5.69	0	0.29	6.29	0.04	26.69	8.09
I	1	5	3.25	0.5	8	5.49	4.5	0.29	0	4	0.25	25	9
J	9	5	3.25	6.5	4	3.49	2.5	6.29	4	0	6.25	17	13
K	0.25	6.25	4.5	0.25	10.25	7.24	6.25	0.04	0.25	6.25	0	28.25	9.25
L	32	8	10.25	24.5	5	7.09	8.5	26.69	25	17	28.25	0	10
M	10	2	3.25	6.5	5	3.69	4.5	8.09	9	13	9.25	10	0

A,D,H,K B,C,F,G B,C,F,G A,D,H,I,K E,F,G B,C,E,F,G B,C,E,F,G A,D,H,I,K D,H,I,K J A,D,H,I,K L M

		Core	Boundary point /Outlier	Boundary point
A	A,D,H,K		Core	
B	B,C,F,G		Core	
C	B,C,F,G		Core	
D	A,D,H,I,K		Core	
E	E,F,H		Boundary /Outlier	Boundary
F	B,C,E,F,G		Core	
G	B,C,E,F,G		Core	
H	A,D,H,I,K		Core	
I	D,H,I,K		Core	
J	J		Outlier	
K	A,D,H,I,K		Core	
L	L		Outlier	
M	M		Outlier	

Condition: minimum point is 4

C-2

C-1

C-1

C-2

C-1

C-1

C-2

C-2

C-2

C-1

C-2

C-1: Cluster 1

C-2: Cluster 2

C-1	B,C,E,F,G	(3,4),(2.5,4), (3,5), (2.8,4.5), (2.5,4.5)
C-2	A,D,H,I,K	(1,2), (1.5,2.5), (1.2,2.5), (1,3), (1,2.5)
Outlier	J,L,M	(1,5), (5,6), (4,3)

Thank you